

Spatial data models

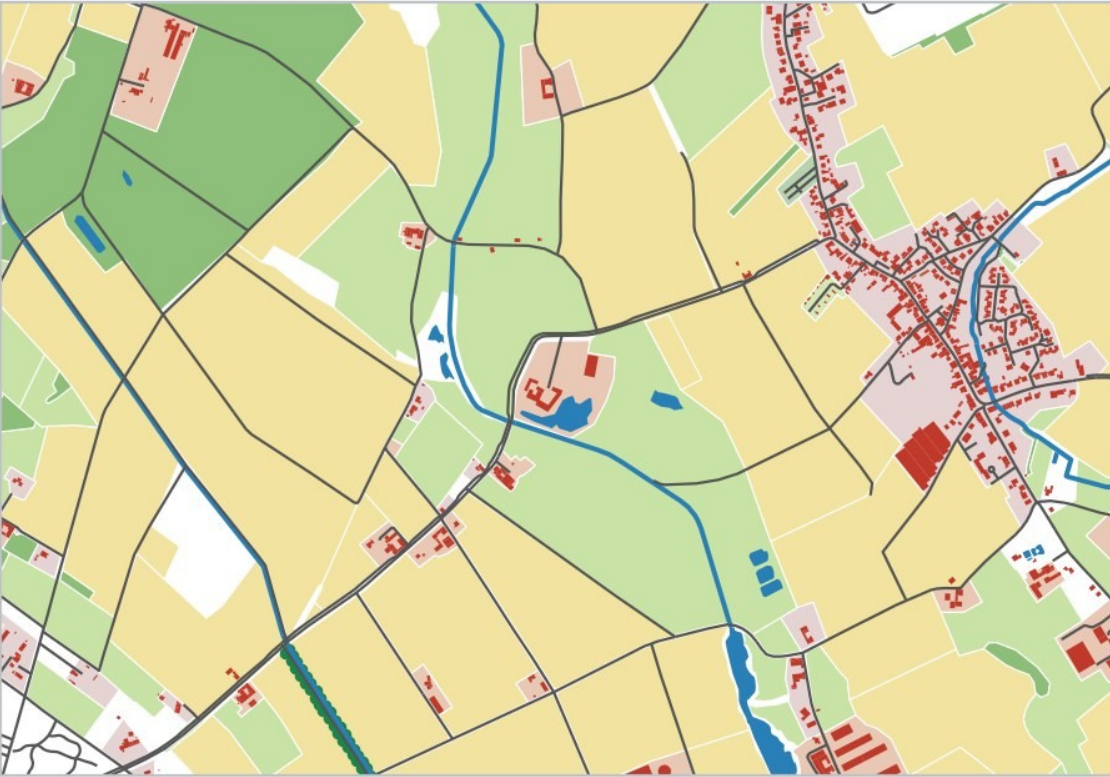
Vector, raster and the OGC Simple Features model

Examples: Kamp-Lintfort and Frohnenbruch



Same place, two models

Vector: discrete objects



Raster: continuous field of 10 m cells

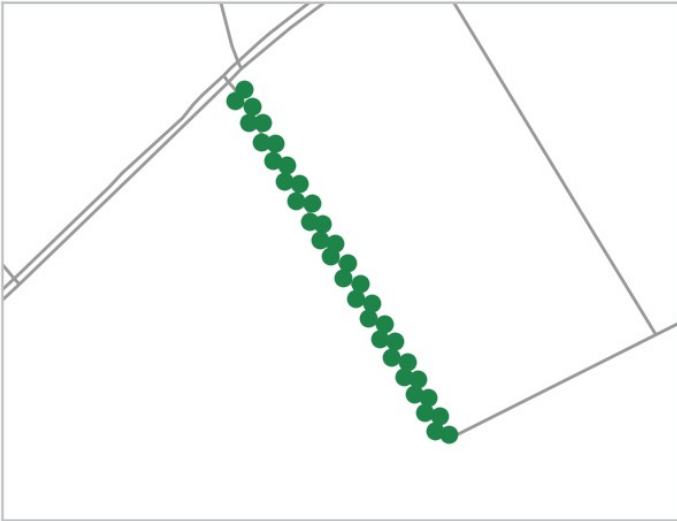


Vector models things that have a boundary and an identity. **Raster** models a quantity that varies everywhere in space.



Vector model: geometry + attributes

Points: trees (n = 37), zoomed



Lines: roads, streams (n = 190)



Polygons: buildings, ponds (n = 902)



fid	building	name	geometry
40	yes	Freie evangelische G...	POLYGON (...)
1	yes		POLYGON (...)

In QGIS

Layer → Add Layer → Add Vector Layer

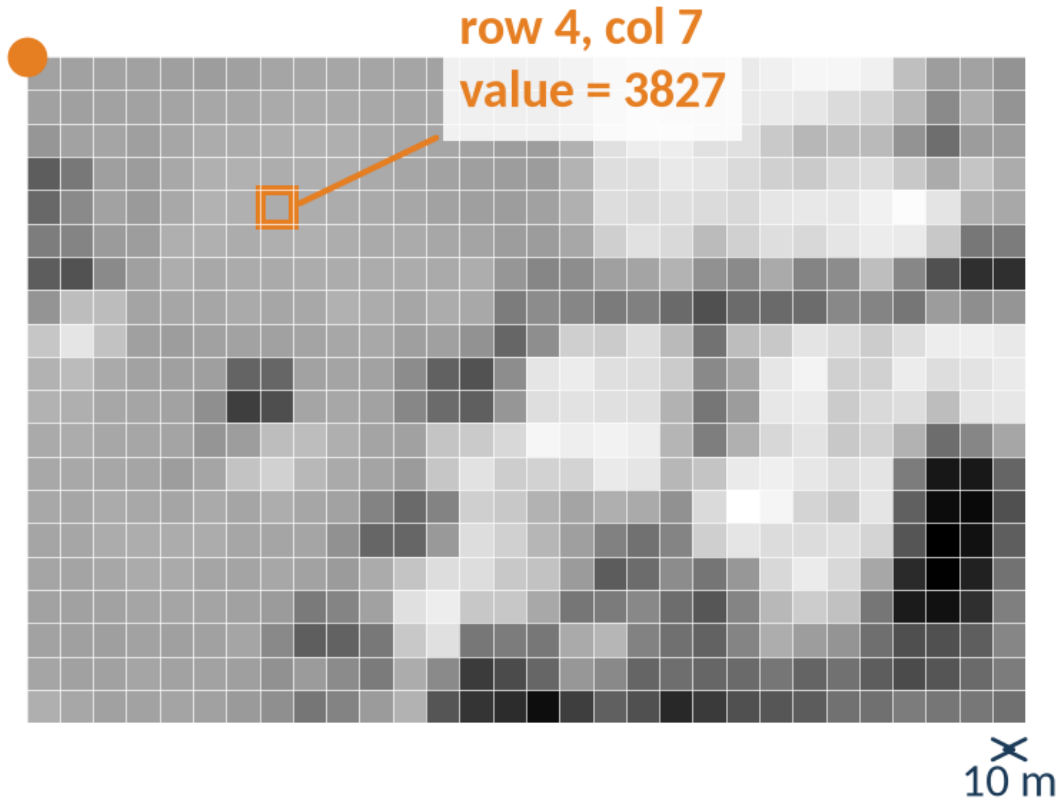
Right-click → Open Attribute Table

Identify Features: click a building



Raster model: a grid of cells

origin (x_0 y_0)
= (323 400 5 709 600)



- **Grid:** rows $\times$ columns of cells (pixels)
- **Cell size:** resolution, here 10 m $\times$ 10 m
- **Bands:** several values per cell (B02 ... B08)
- **NoData:** a value that means 'no measurement'
- **CRS + origin:** place the grid on the Earth

In QGIS

Layer Properties $\rightarrow$ Information:

size, pixel size, CRS, bands, NoData

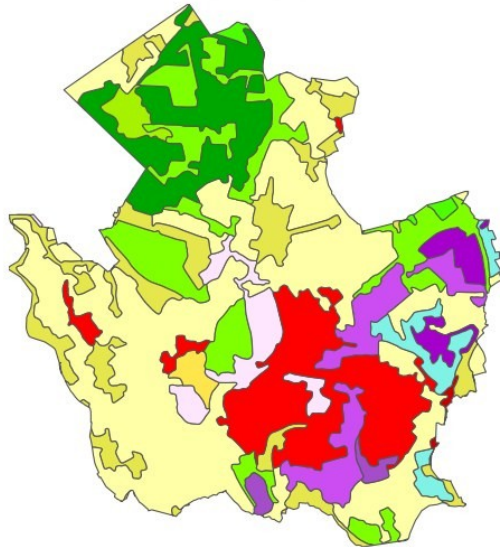
Identify Features: values of a pixel

Zoom in until you see the cells



Resolution decides what you can see

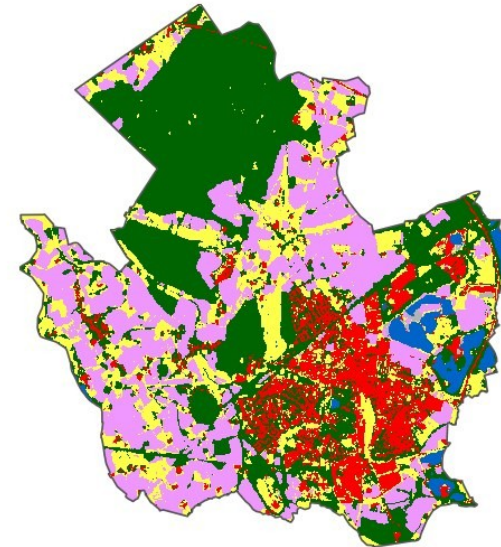
CORINE 2018: 72 polygons, MMU 25 ha



- 231 Pastures
- 211 Arable land
- 311 Broad-leaved forest
- 112 Discontinuous urban

- 142 Sport/leisure
- 121 Industrial/commercial
- 512 Water bodies
- 131 Mineral extraction

ESA WorldCover 2021: 631 512 cells of 10 m



- Tree cover
- Cropland
- Water
- Shrubland
- Built-up
- Wetland
- Grassland
- Bare/sparse

CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA · ESA WorldCover 10 m 2021 v200 (CC BY 4.0)



Vector or raster?

	Vector	Raster
Models	discrete objects with identity	continuous fields, images
Geometry	points, lines, polygons (exact coordinates)	regular grid, cell size = resolution
Attributes	many columns per feature	one value per cell and band
Strong at	topology, networks, precise boundaries, cartography	map algebra, surfaces, remote sensing
Weak at	continuous variation	sharp boundaries, storage of sparse data
Formats	GeoPackage, GeoJSON, Shapefile, PostGIS, FlatGeobuf	GeoTIFF / COG, NetCDF, Zarr, JPEG2000
Python	shapely, GeoPandas	rasterio, xarray / rioxarray



File formats you will meet

Format	Model	Files	Good to know
GeoPackage (.gpkg)	vector + raster	1 (SQLite)	many layers, long field names, OGC standard: our default
Shapefile (.shp)	vector	3-7 (.shp .shx .dbf .prj ...)	one layer, field names $\leq$ 10 characters, 2 GB limit
GeoJSON (.geojson)	vector	1 (text)	web, always WGS 84 lon/lat (RFC 7946), large files get slow
CSV with x/y	points only	1 (text)	no CRS stored: you must know it; keys as text (leading zeros)
GeoTIFF / COG (.tif)	raster	1	bands + georeference; COG = cloud-optimised, read in pieces
NetCDF / Zarr	raster cubes	1 / folder	time series of grids (climate, RADOLAN)

Rule of thumb: GeoPackage for vector exchange, GeoTIFF/COG for rasters, PostGIS when many people work on the same data (geo0840).



Attribute types matter

Type	Example	Pitfall
text (string)	name = 'Haus Frohnenbruch', KN = '05170020'	keys stay text
integer	building:levels = 2	no decimals
decimal (real)	area_ha = 6315.2	unit in the field name
date	survey_date = 2025-05-13	CSV import may read text
boolean	active = true	Shapefile has none

'05170020' → 5170020
a key read as a number loses its leading zero

In QGIS

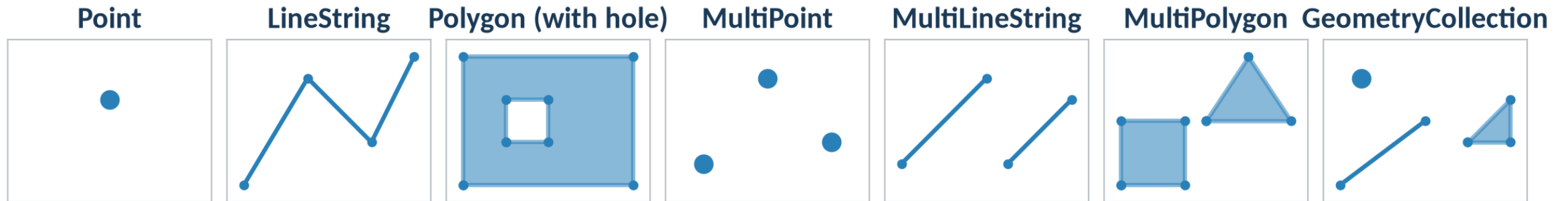
Layer Properties → Fields

CSV import: untick 'Detect field types' for keys

Field Calculator: to_string(), to_int()



Seven geometry types, written as WKT



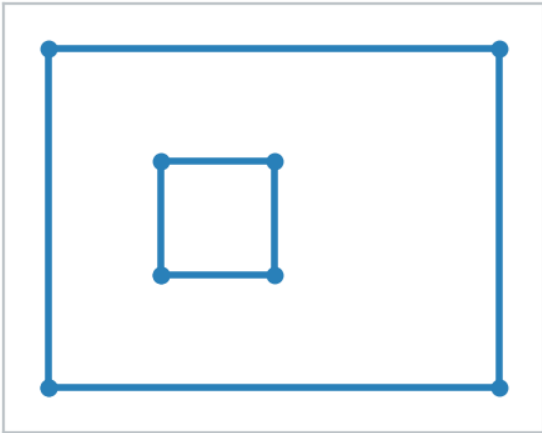
```
POINT (2 2)
LINESTRING (0 0, 1.5 2.5, 3 1, 4 3)
POLYGON ((0 0, 4 0, 4 3, 0 3, 0 0), (1 1, 2 1, 2 2, 1 2, 1 1))
MULTIPOINT ((0.5 0.5), (2 2.5), (3.5 1))
MULTILINESTRING ((0 0.5, 2 2.5), (2.5 0.5, 4 2))
MULTIPOLYGON (((0 0, 1.5 0, 1.5 1.5, 0 1.5, 0 0)), ((2 1.5, 4 1.5, 3 3, 2 1.5)))
GEOMETRYCOLLECTION (POINT (0.5 2.5), LINESTRING (0 0, 2 1.5), POLYGON ((3 1, 4 1, 4 2, 3 1)))
```

```
g = shapely.wkt.loads("POINT (2 2)"); g.wkt, g.wkb_hex
```

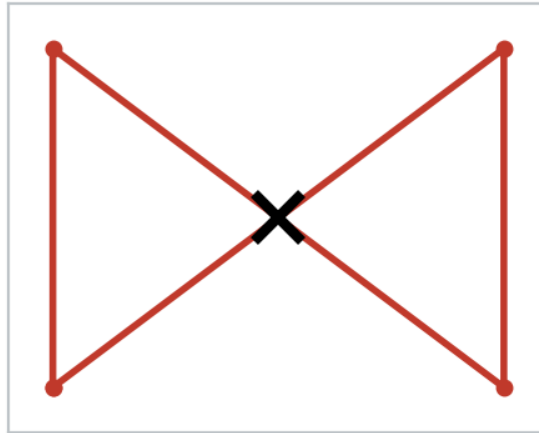


Valid geometries: rules of the model

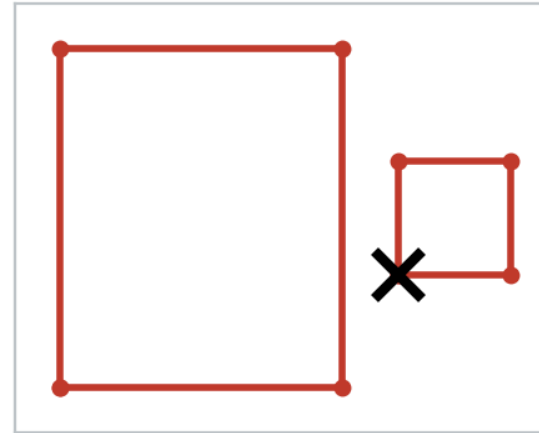
✓ valid



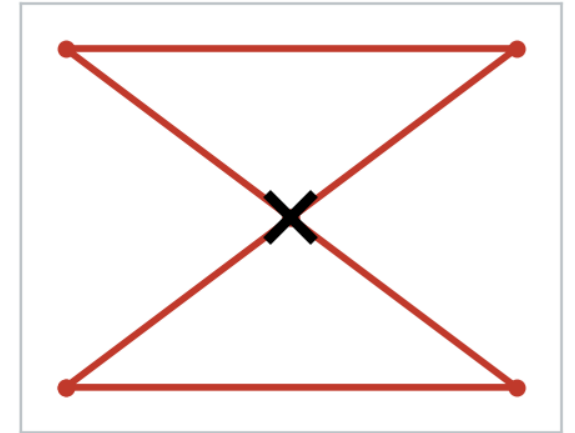
✗ self-intersection



✗ hole outside shell



✗ ring self-touch



- Rings are closed and do not cross themselves
- Holes lie inside the shell, touching it in one point at most
- Parts of a MultiPolygon do not overlap

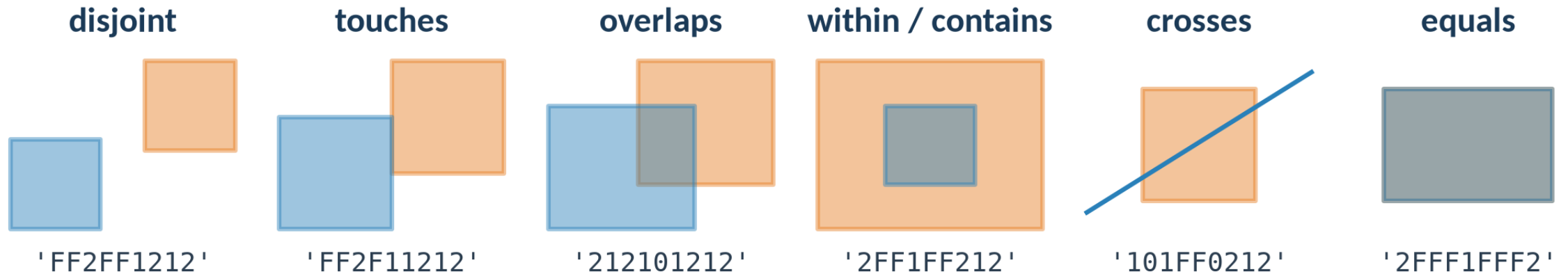
In QGIS

Vector → Geometry Tools → Check Validity

Processing → Fix geometries



Spatial relationships between objects



Relationship	means
disjoint	no common point
touches	common boundary, no common interior
overlaps / crosses	share part of the interior
within / contains	one lies completely inside the other
intersects	anything but disjoint

In QGIS

Vector → Research Tools →
Select by Location
Processing → Join attributes
by location (within, ...)



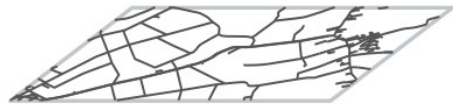
Thinking in layers



trees



buildings



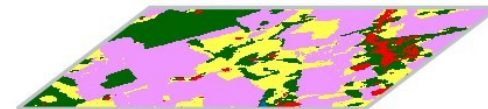
roads



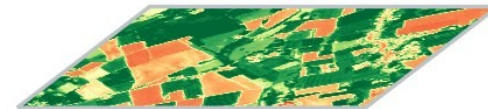
water



landuse



land cover



NDVI



B08 NIR



B04 red

Vector stack: overlay geometries

Raster stack: combine aligned cells

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geo0160 · Spatial data models · opengeo



Take-aways

1

Two models

Vector for objects, raster for fields; the question and the scale decide.

2

Formats and types

GeoPackage and GeoTIFF; keys as text, know the CRS of a CSV.

3

Valid geometries + relationships

Check validity before analysis; select and join by location.

4

Layers

Same CRS, same extent: the precondition for overlay and map algebra.



Sources and further reading

- **Standards:** OGC 06-103r4 Simple Feature Access Part 1 (ISO 19125-1); OGC GeoPackage 1.4; DE-9IM: Clementini, Di Felice & van Oosterom (1993)
- **Data:** © OpenStreetMap contributors (ODbL); Contains modified Copernicus Sentinel-2 data 2025, scene S2A_32ULC_20250513 via Earth Search STAC; CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA; ESA WorldCover 10 m 2021 v200 (CC BY 4.0)
- **Further reading (links):** Geo-Python and Automating GIS Processes (Univ. of Helsinki); pythongis.org (Tenkanen, Heikinheimo, Whipp); Shapely and GeoPandas user guides; QGIS Training Manual – vector and raster chapters
- **Figures:** own work, generated with shapely, GeoPandas, rasterio and matplotlib (make_slides.py)

